

TrustLayer AI: A Multimodal Deepfake Detection Framework with an Agentic Reasoning Layer for Securing Corporate Video Communication

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Abstract

Generative AI has created a new cyber-threat – deepfake-enabled phishing, where attackers synthesise executives in real time over video conferencing to approve fraudulent transfers. Losses have been documented – most notably US\$25 million in an incident in Hong Kong in 2024 – showing this to be an active vector for financial crime. Traditional security controls do not semantically understand the authenticity of audio-visuals. In this work, we propose TrustLayer AI, a multimodal deepfake detection framework with an added agentic reasoning layer. The system combines three complementary signals: (i) a visual classifier based on CNN, (ii) a pretrained audio anti-spoofing model, and (iii) a cross-modal consistency module that links the motion of lip-landmarks and the audio amplitude. An agentic layer applies business context (meeting type, requested action, score trends, after-hours status, first-meeting flags) to make threat-appropriate decisions (allow, soft-warn, escalate, hard-block) and provide natural-language explanations and audit logs. Performance of the TrustLayer fusion on FakeAVCeleb v1.2 is calibrated balanced accuracy of 0.620 and F1 of 0.735. The agentic layer is evaluated through a 24-cell scenario simulation that exhibits context-sensitive decisions from allow on routine calls to hard-block on high-stakes finance requests.

Index Terms — *deepfake detection, multimodal fusion, agentic AI, corporate cybersecurity, cross-modal consistency, video conferencing security*

I. INTRODUCTION

Corporate communication has shifted decisively to video conferencing platforms and generative AI has matured in parallel to enable photorealistic face synthesis [1], face swapping pipelines [2], audio-driven facial reenactment [3], and five-second voice cloning [4]. Together these capabilities enable deepfake-enabled social engineering in live calls.

In 2024, a Hong Kong incident led to a US\$25 million wire-transfer fraud when an employee attended a video conference with a synthesised CFO and colleagues [5]. Similar incidents have been reported in the financial service, energy and technology sectors [6]–[8]. The FBI and the UK NCSC have issued official warnings [9], [10]. Traditional security tools, however – firewalls, MFA, EDR, DLP – operate at the network or content-metadata level and have no semantic awareness of whether a face or voice is real [11].

On high-quality deepfakes, human detection performance is 60-70% – just above chance [12]. Biometric verification is itself vulnerable to the same generative capabilities it is intended to prevent [13]. The verification gap is apparent: there is no tool in use that flags synthetic participants in a live call.

In this paper, we introduce TrustLayer AI, a multimodal deepfake detection framework enhanced with an agentic reasoning layer. Our main contributions are: (1) a fused visual + audio + cross-modal detection pipeline; (2) an agentic layer that translates imperfect detector signals + business context into operational decisions; (3) explanations in natural language for non-technical end users; and (4) per-

IV. AGENTIC REASONING LAYER

The main contribution is the agentic reasoning layer. It takes the fused trust score and business metadata and produces a composite risk score, an operational action, a natural-language explanation, and a structured audit log. Deterministic rule core guarantees reproducibility; optional LLM augmentation (via Anthropic’s Claude API) generates rich context explanations.

A. Risk Computation

Five pieces are combined: (1) base detector risk (1 - trust score); (2) meeting type multiplier (finance x1.20, external x1.15, board x1.15, HR x1.05, internal x1.00); (3) action-keyword sensitivity (+0.10 for each sensitive keyword like "wire", "password", "classified", max 0.25); (4) trust trend (+0.10 if mean successive score difference < -0.10); (5) contextual red flags (+0.08 for first-meeting on high-stakes call; +0.07 for after-hours finance). The composite is clipped to $[0,1]$.

B. Action Selection

Risk < 0.30 -> ALLOW; [0.30, 0.70) -> SOFT_WARN; [0.70, 0.85) -> ESCALATE; ≥ 0.85 -> HARD_BLOCK. The four discrete actions rather than a continuous score reflect the way operational security tools are used: end users need a clear directive rather than a probability.

C. Explanation and Audit Logging

The deterministic layer decides the action, the LLM (if available) translates it into prose suitable to the non technical employees in context. Each decision is serialised to JSON for SIEM ingestion including timestamp, participant claim, detector outputs, risk score, action and explanation.

category evaluation on FakeAVCeleb v1.2 for all four real-fake combinations.

II. RELATED WORK

A. Visual Deepfake Detection

The visual detection has gone through three generations: hand-crafted artefact detectors [14], CNN-based classifiers on FaceForensics++ [15] and vision-transformer hybrids [16]. The accuracy on in-domain benchmarks is above 95 %, but domain shift, especially due to video compression or novel generators, degrades it significantly [17].

B. Audio Anti-Spoofing

The audio anti-spoofing has been formalised since 2015 through the ASVspoof challenge series [18]. MFCCs [19] and CQCCs [20] are still strong baselines. End-to-end learned models (RawNet2 [21], AASIST [22]) are now on par with or better than hand-crafted features, with sufficient training data. The self-supervised pre-training of wav2vec 2.0 results in state-of-the-art out-of-domain performance [23].

C. Multimodal and Cross-Modal Detection

When a single modality is manipulated, unimodal detectors fail in a predictable manner. Modality Dissonance Score [24] leverages learned semantic similarity across modalities; emotion-consistency methods [25] utilise affective cues; Khalid and Woo [26] learn joint audio-visual embeddings on FakeAVCeleb. The canonical cross-modal synchrony metric is SyncNet [27] adversarially trained against Wav2Lip [3]. Korshunov and Marcel [28] show that simpler lip-sync proxies do not generalise to unseen generators.

D. Agentic AI for Security

LLM based agentic systems have been used for penetration testing (PentestGPT [29]) and intrusion detection. The ReAct [30] and Reflexion [31] frameworks exhibit interleaved reasoning and action. This work is motivated by the open problem of applying agentic reasoning to live-call deepfake security.

III. SYSTEM ARCHITECTURE

TrustLayer AI is a pipeline that has five components: sensing and then reasoning.

A. Visual Detection Branch

We sample 8 frames per clip equally spaced. We detect faces with OpenCV Haar cascade (20% margin) and crop to 224×224 pixels. The pretrained model prithivMLmods/Deep-Fake-Detector-v2-Model (Hugging Face) outputs a per-frame P(fake); per-clip prediction is the arithmetic mean across frames. This is a task well suited to CNNs, since face-swap artefacts (jawline warping, skin-texture inconsistency, frequency anomalies) are distributional outliers in pixel space.

B. Audio Detection Branch

V. EXPERIMENTAL EVALUATION

A. Dataset and Setup

FakeAVCeleb v1.2 [26] contains 20,000+ videos of four real-fake categories RVRA (real video, real audio), RVFA (real video, cloned audio), FVRA (face-swap, real audio), FVFA (face-swap, cloned audio) generated using FaceSwap, FSGAN, Wav2Lip and SV2TTS. We selected a balanced subset of 200 videos (50 per category) by subject-disjoint sampling with a fixed seed of 42. We used a 50/50 train-validation split, yielding 100 videos for threshold tuning and 100 for held-out testing. Compute: Python 3.11, PyTorch, Hugging Face Transformers, librosa, MediaPipe, Google Colab T4 GPU.

B. Overall Detection Performance

TABLE I: Overall Detection Performance

Method	Bal Acc	F1	ROC-AUC
M1: Visual-only	0.528	0.632	0.553
M2: Audio-only	0.539	0.406	0.654
M3: Late Fusion	0.562	0.668	0.570
M4: TrustLayer	0.620	0.735	0.572

M4 TrustLayer achieves the best calibrated balanced accuracy (0.620) and F1 (0.735). M2 Audio-only achieves the best threshold-independent ROC-AUC (0.654), indicating more absolute discriminative signal in the audio model on this dataset. The PR-AUC values are consistent across methods (0.787-0.803), which suggests that all methods broadly rank fake examples correctly.

C. Per-Category Performance

TABLE II: Per-Category Balanced Accuracy

Method	RVRA	RVFA	FVRA	FVFA
M1: Visual	0.520	0.385	0.607	0.624
M2: Audio	0.807	0.314	0.262	0.235
M3: Fusion	0.547	0.513	0.600	0.624
M4: TrustLayer	0.367	0.622	0.738	0.758

The per-category breakdown demonstrates the basic motivation of multimodal fusion. M2 Audio-only achieves 0.807 on RVRA but deteriorates on visual-fake categories (0.24-0.31). M1 Visual-only is the reverse: strong on FVRA/FVFA but weak on RVFA (0.385). M4 TrustLayer trades RVRA accuracy (0.367) for much better coverage of all three fake categories, which is the right operational trade-off for a security tool.

D. Cross-Modal Sync Diagnostic

The landmark-based sync implementation achieves an average separation between real and fake of 0.056 (real mean 0.524, fake mean 0.468, SD ~ 0.132 both). This leads to a better operating point for M4 than M3. The signal still remains weaker than the unimodal detectors, which is consistent with

I extract the audio from the .mp4 container with librosa, resample to 16 kHz mono and pass it to MelodyMachine / Deepfake-audio-detection-V2 (Hugging Face). Anti-spoofing models exploit spectral signatures in synthetic speech: smoothed formant transitions, atypical cepstral energy and reduced micro-prosodic variability. As a sanity check, a custom classifier based on MFCC was implemented, which was outperformed by the pretrained model on the held-out split.

C. Cross-Modal Consistency Branch

In real video, there is a close temporal correspondence between the visible articulation and the produced phonemes. In the improved landmark based implementation, MediaPipe FaceMesh is used to extract lip landmarks (13, 14, 78, 308) and then computes the normalised lip-distance ratio and Pearson correlation with the librosa RMS amplitude envelope. The output is scaled to $[0, 1]$. A first pixel-motion proxy yielded no meaningful separation (means 0.488 vs 0.493). The landmark-based version yields a real-vs-fake mean separation of 0.056.

D. Fusion Strategies

We experimented four fusion strategies, M1 (visual only), M2 (audio only), M3 (late fusion at equal weights: $0.5 \cdot P_v + 0.5 \cdot P_a$), and M4 TrustLayer ($0.4 \cdot P_v + 0.4 \cdot P_a + 0.2 \cdot (1 - \text{sync})$). Threshold selection for each method uses balanced accuracy on a held-out validation split, replacing an earlier F1-maximisation approach that produced degenerate always-fake predictions under class imbalance.

Wav2Lip being adversarially trained against the lip-sync metrics.

E. Agentic Layer Scenario Simulation

TABLE III: Agent Decision Matrix (selected cells)

Scenario	RVRA	FVFA
CFO wire transfer (after-hours)	soft_warn (0.45)	hard_block (1.00)
Routine internal stand-up	allow (0.15)	hard_block (0.85)
Board classified (after-hours)	soft_warn (0.37)	hard_block (1.00)
HR onboarding	allow (0.16)	hard_block (0.89)

The 24-cell scenario matrix illuminates three properties that distinguish an agentic tool from a raw classifier: (1) the same detector input can lead to different actions depending on context — an RVRA signal leads to soft-warn on a CFO wire transfer but allow on a routine stand-up; (2) the same context can lead to different actions depending on detector input; (3) low-stakes calls are given the benefit of the doubt even if individual modality detectors fire.

VI. DISCUSSION

A. Detector Performance and Limitations

Absolute ROC-AUC scores (0.55-0.65) are below deployment stand-alone thresholds, due to distributional mismatch between off-the-shelf pretrained models and specific generation pipelines of FakeAVCeleb. Three levers of improvement would push performance to the 0.85-0.90 range: in-domain fine-tuning (+0.15-0.25 ROC-AUC), SyncNet integration for cross-modal signal, and learned fusion instead of fixed weights (+0.05-0.12 ROC-AUC). They each need roughly an order of magnitude more compute than the T4 educational-tier used here.

B. Agentic Layer as Primary Defence

The modest detection accuracy does not weaken the thesis of the paper, but rather strengthens it: when no single detector is reliable, the main practical defence is contextual reasoning over multiple weak signals, along with business context. The end-to-end demonstration (Section B.3 of the full thesis) shows that a SOFT_WARN generated for a high-stakes call where the fused detector reported trust 0.828 was the correct outcome — a thresholded classifier would have accepted the call outright.

VII. CONCLUSION

In this work we introduced TrustLayer AI, a multi-modal deepfake detection framework with an agentic reasoning layer for securing corporate video communication. The system uses CNN visual, pretrained audio anti-spoofing, and landmark-based cross-modal signals to output a properly calibrated trust score, and an agentic layer to combine this with business context to generate operationally appropriate decisions and explanations. The M4 TrustLayer fusion attains a calibrated

balanced accuracy of 0.620 and F1 of 0.735 on FakeAVCeleb v1.2. The 24-cell scenario simulation depicts context-sensitive decisions across a range of corporate threat scenarios.

Future work includes in-domain fine-tuning and SyncNet integration to lift detection accuracy, real-time WebRTC stream integration, adversarial robustness evaluation, controlled user studies on LLM-generated explanations, and federated learning for organisation-specific policy adaptation.

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